3D MODEL

View online: https://www.construct.net/en/make-games/manuals/construct-3/plugin-reference/3d-model

The **3D model** plugin can be used to display models from a **.gltf** or **.glb** file in the Layout View, along with playing their animations.

Check the 3D model manual entry for information about how to import model files.

Refer to the 3D shape manual entry for information about working with 3D in Construct 3 which is also relevant to this plugin.

Scripting

When using JavaScript or TypeScript coding, the features of this object can be accessed via the I3DModelInstance script interface.

Collisions

Similarly to 3D Shape collisions for 3D models still happen in 2D.

3D model properties

3D Model

The 3D model object the instance will use.

Initially visible

If the model should be visible or not when the layout starts.

X offset

The offset in the X axis in relation to the corresponding instance.

Y offset

The offset in the Y axis in relation to the corresponding instance.

Z offset

The offset in the Z axis in relation to the corresponding instance.

X scale

The scale in the X axis. This is compounded with the scale of the corresponding instance.

Y scale

The scale in the Y axis. This is compounded with the scale of the corresponding instance.

Z scale

The scale in the Z axis. This is compounded with the scale of the corresponding instance.

X origin

The transformation origin in the X axis.

Y origin

The transformation origin in the Y axis.

Z origin

The transformation origin in the Z axis.

Transform mode

How the model transforms when the corresponding instance is resized. *"Fit"* mode preserves the model's proportions. *"Stretch"* mode scales each axis to fit in the instance bounds. *"Legacy"* mode exists for backwards compatibility, the other modes are preferred.

Meshes

Toggle which meshes to show.

Animation

The initial animation.

Loop

Check so they animation loops.

Mesh render mode

The mesh render mode. Can be *"hierarchy"* to draw meshes using the hierarchy defined in the model file, *"isolate"* will draw the enabled models as if they where the only ones in the model file.

Initially playing

If the model should start playing an animation or not when the layout starts.

Initial animation progress

The initial animation progress.

Enable collisions

If the instance should collide or not.

Back face culling

Enable or disable back face culling.

Bounding box

An editor only property to display the bounding box of the 3D model.

Bounding box color

An editor only property to change the color of the bounding box of the 3D model.

3D model conditions

For conditions in common to other objects, see Common conditions.

On model loaded

Triggered when a new 3D model object is loaded.

On model load error

Triggered when there is an error loading a 3D model object.

For each mesh

Loop through all the meshes in the 3D model.

For each animation

Loop through all the animations in the 3D model.

Is playing

Checks if there is an animation playing.

Is looping

Checks if animations are set to loop.

Render mode

The render mode indicates how the meshes are displayed. If *"hierarchy"* is used, all the displayed meshes are drawn following the authored hierarchy in the model file. If *"isolate"* is used, the displayed meshes are drawn as if they where the only ones in the model file.

Is mesh enabled

Checks if the provided mesh name is being drawn or not.

Are all meshes enabled

Checks if all the meshes in the model are being drawn.

Mesh exists

Checks if a mesh exists in the model.

Enable collisions

Checks if collisions are enabled.

Is back face culling enabled

Checks if back face culling is enabled.

On animation finished

Triggers when the provided animation finishes.

On animation looped

Triggers when the provided animation loops.

On any animation finished

Triggers when any animation finishes.

On any animation looped

Triggers when any animation loops.

On texture URL loaded

Triggered after 'Load texture from URL' when the texture has finished loading.

On texture URL failed

Triggered after 'Load texture from URL' if the texture fails to load.

On all textures released

Triggered after unload dynamic textures completes.

3D model actions

For actions common to other objects, see Common actions.

Set 3D model

Start loading a new 3D model object. The **"Mesh"** parameter can optionally be used to draw a single mesh, if the parameter is left empty all the meshes will be drawn. Use the **"Animation"** parameter to choose the starting animation, if left empty the first animation is used. Choose if the animation should start playing using the **"Playing"** parameter, use the **"Loop"** parameter to choose if the animation should loop and the **"Progress"** parameter to choose where the animation should start from, using a value in the range **[0 - 1]**.

Set 3D model (by name)

This is the same as **"Set 3D model"** but the 3D model object is specified using a string.

Set mesh enabled

Choose if a mesh is drawn or hidden.

Set all meshes enabled

Choose if all meshes are drawn or hidden.

Set animation

Change the current animation. Choose if the animation should play using the **"Playing"** parameter, use the **"Loop"** parameter to choose if the animation should loop and use the **"Progress"** parameter to choose where the animation should start from, using a value in the range **[0 - 1]**.

Set looping

Choose if animations should loop or not.

Set progress

Set the progress of the current animation using a value in the range **[0 - 1]**.

Set origin

Set the transformation origin on all three axes.

Set transform

Change the transformation values of the model. **"Transform X"**, **"Transform Y"**, **"Transform Z"** correspond to each axis. Use the **"Operation"** parameter to choose which operation to do on the values, either *Set*, *Add*, *Subtract*, *Multiply* or *Divide*. Use the **"Type"** parameter to choose which transformation to affect, either *Offset*, *Rotation* or *Scale*.

Play

Start playing the specified animation, use the **"Loop"** parameter to choose if the animation should loop and use the **"Progress"** parameter to choose where the animation should start from, using a value in the range **[0 - 1]**.

Stop

Stop the current animation. The progress is reset to 0.

Pause

Pause the current animation. The progress stays as is.

Resume

Resume a paused animation.

Set mesh render mode

Change the render mode of the model. Either *"hierarchy"* or *"isolate"*.

Set collisions enabled

Change the collision state.

Set back face culling

Change the state of back face culling.

Load texture from URL

Replace the currently loaded texture into the specified mesh. Use the *"Target"* parameter to choose which instances are affected, "Object type" will affect all the instances of the picked object type that are using the same 3D model, "Picked instances" will only affected the picked instances in the current action block that are using the same 3D model.

Unload all dynamic textures

Unload all dynamically loaded textures.

Unload some dynamic textures

Unload dynamically loaded textures from a 3D model's mesh.

Unload some dynamic textures (by name)

Unload dynamically loaded textures from a 3D model's mesh, providing the 3D model as a string name.

3D model expressions

For expressions common to other objects, see common expressions.

Error3dModelName

The name of the last 3D model object that failed to load.

Name

The name of the 3D model object.

Meshes

A comma separated string with all the currently enabled mesh names.

Animation

The current animation.

AnimationDuration(Animation)

The total duration of the specified animation.

Progress

The progress of the current animation as a value in the range **[0 - 1]**.

MeshCount

The amount of meshes in the current model.

MeshAt(Index)

The mesh at the specified index.

AnimationCount

The total amount of animations in the model.

AnimationAt(Index)

The animation at the specified index.

CurMesh

The current mesh in the **"For each mesh"** looping condition.

CurAnimation

The current animation in the **"For each animation"** looping condition.

OffsetX

The offset in the X axis in relation to the corresponding instance.

OffsetY

The offset in the Y axis in relation to the corresponding instance.

OffsetZ

The offset in the Z axis in relation to the corresponding instance.

ScaleX

The scale in the X axis. This is compounded with the scale of the corresponding instance.

ScaleY

The scale in the Y axis. This is compounded with the scale of the corresponding instance.

ScaleZ

The scale in the Z axis. This is compounded with the scale of the corresponding instance.

OriginX

The transformation origin in the X axis.

OriginY

The transformation origin in the Y axis.

OriginZ

The transformation origin in the Z axis.